

ARUN KUMAR GADDAM

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CAREER OBJECTIVE:

To obtain a responsible and challenging Embedded Software Professional position where my education and work experience will have valuable application.

PROFESSIONAL SUMMARY:

- Embedded Software Professional with 4+ years of experience in developing and debugging embedded C code, understanding existing projects and providing software support, unit level testing, Integration and Qualification testing of Embedded Software. Self- motivated and ability to pursue task completion with minimum supervision.
- Currently associate as Senior Engineer with **MAGNA AUTOMOTIVE INDIA (MPT)**
- ISTQB CTFL Certified Test Engineer.

TECHNICAL EXPERTISE:

- ❖ **Programming Languages:** -
 - C language
 - Embedded C
 - C++ language
 - Python
 - CAPL (Communication access programming language)
- ❖ **Development, Design and Debugging Tools:** -
 - **Software tools** – Dspace Control Desk, ECU Test Tool, CANoe, Cantata, Trace32, VFlash, Silver.
 - **Communication Interfaces** - I2C, CAN.
 - **Version control tools** – SVN Tortoise, Git
 - **Application Management** – PTC IALM, CodeBeamer ALM, IBM Jazz

EMPLOYMENT SUMMARY:

- ❖ **Current Employer:** **MAGNA AUTOMOTIVE INDIA** [Jan'2025 – Present]
Designation:
 - **Senior Engineer (Jan'2025 – Present)**

Working as HIL Test Engineer for eDrive (EV Powertrain) Inverter SW team that works on Design, Development, Validation and Verification for Mercedes and BMW OEMs with compliance to AUTOSAR, ASPICE and ISO 26262 Functional safety standards.

❖ **Past Employer: TATAELXSI LTD [Feb'2021 – Dec2024]**
Designation:

- **Senior Engineer (Feb'2021 – Present)**

Worked as Embedded Senior Software Engineer in Automotive team that works on Design, Development, Validation and Verification of Software module with compliance to AUTOSAR, ASPICE and ISO 26262 standards.

PROJECT PROFILES:

PROJECT #1

Project : HIL Test for eDrive Inverter SW (Safety & IoHwAb)
Parent Company : MAGNA Automotive India.
Designation : Engineer Testing
Client Company : BMW, Daimler OEM's
Environment : Dspace Scalexio & Control Desk, ECU Test Tool, Python Framework, PTC IALM, GitHub.

Project Description : Hardware In Loop testing for eDrive (EV powertrain) inverter software components with Dspace Scalexio, Control Desk, Python XIL API's Framework and ECU Test Tool Environment.

Rotor Position Sensor, IO Powerstage, IO ECU, Startup tests, Safety Powertrain monitor and Safety Fault Manager are Software Components Responsible for Functional Testing.

Roles and Responsibilities:

- Working on Functional and Fault Matrix topics for Safety and IoHwAb software components HIL testing.
- Analysis of Software Requirements to derive test specification for Safety and IoHwAb Inverter Software Application Components.
- Preparation of Software Qualification test specification in PTC IALM.
- Testscript Implementation using Python Framework and API's
- Testscript Implementation in ECU Test Packages.
- Test Execution using Dspace Scalexio, Control Desk, ECU Test, Python Framework
- Generation of test report & Bug reporting.
- Confirmation and Regression testing.

PROJECT #2

Project : SIL Test for eDrive Inverter Software
Parent Company : TataElxsi Ltd, Bengaluru
Designation : Senior Engineer
Environment : Synopsys Silver, Python Framework, IBM Jazz
Client Company : MAGNA Automotive India.

Project Description : Software In Loop Interface testing for eDrive inverter software components using Synopsys Silver Environment with enhanced test automation python framework.

Roles and Responsibilities:

- Analysis of Software Interface Requirements to perform Qualification testing of Inverter software components.
- Preparation of Software Qualification test specification.
- Software Interface Qualification test of **Signal Input/Signal Output** components of eDrive Inverter SW with Synopsys Silver and python Framework
- Generation of test report & Bug reporting.
- Confirmation and Regression testing.

PROJECT #3

Project : Bootloader Porting and Testing
Parent Company : Tata Elxsi Ltd, Bengaluru
Designation : Senior Engineer
Environment : CANoe, CAPL, VFlash, JIRA
Client Company : Nexteer Automotive

Project Description : Bootloader Software porting by customizing existing Bootloader SW of one Hardware architecture to support another Hardware architecture. Qualification testing of Bootloader Software with CANoe, CAPL and UDS. Flashing of Application Software after successful customization using CAPL developed Flash script.

Roles and Responsibilities:

- Analysis of Software Requirements to perform Qualification testing of Bootloader software components.
- Preparation of Software Qualification test plan compliant to ISO26262 standard.
- Preparation of Software Qualification test specification and ensure that the test cases are compliant to the ISO26262 standard.
- Preparation of High-Level Design (HLD) for Bootloader SW.
- Flashing of Application SW with Bootloader using CANoe and CAPL.
- Software Qualification testing of Bootloader SW with CANoe and CAPL using UDS.
- Generation of test report & Bug reporting.

PROJECT #4

Project : ADAS MCU Software Testing (Parking Control).
Parent Company : Tata Elxsi Ltd, Bengaluru
Designation : Engineer
Environment : Aurix TC39x, CodeBeamer ALM, CANTATA, Trace32, CANoe, JIRA, Gitlab
Client Company : VinFast Automotive (OEM)

Project Description : Software Testing of ADAS Safety MCU equipped with the Infineon controller. SW Unit, Functional, Integration and Qualification (SWE.4, SWE.5, SWE.6) testing of application & Complex Device Drivers for Camera and temperature sensor software components.

Roles and Responsibilities:

- Analysis of the Software Detail Design, Preparation of Unit Test specification and

perform SW Unit testing (SWE.4) of various application layer & CDD software components included under the ADAS ECU MCU using CANTATA and ensure that the test cases are compliant to the ISO26262 (ASIL -D) standard.

- Preparation of Functional test specification & Perform Functional testing of various software components using CANTATA.
- Understanding of Software Architectural Design, Preparation of SW Integration test specification and perform SW Integration testing (SWE.5) of various application layer software components using Trace32, CANoe & CAPL.
- Understanding of Software Requirement specification and Safety requirement specification, Preparation of SW Qualification test specification and perform SW Qualification testing (SWE.6) using Trace32, CANoe & CAPL.
- Perform peer review of Unit test, Functional test, Integration test and Qualification test artifacts of each base practice.
- Mentoring trainee Engineers.
- Discuss project status with the client daily.
- Follow Agile based software development workflow.

PROJECT # 5

Project : ARCAM (Augmented Reality Camera) Algorithm - ADAS
Parent Company : TataElxsi Ltd, Bengaluru
Designation : Engineer
Environment : Enterprise Architect, OpenGL (Graphics Library), JIRA
Client Company : FICOSA - BMW

Project Description : Project that involved in developing a software application to integrate into AR camera application for customer specific head unit. Rendering of video pipeline using OpenGL & distortion correction of fisheye image.

Roles and Responsibilities:

- Developed video pipeline rendering feature using OpenGL graphics library. (Focused on Reducing memory utilization and image frame rendering time).
- Preparation of Software detailed design.
- Preparation of test specification for unit testing of ARCAM application SW.

EDUCATIONAL QUALIFICATIONS:

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|---|-------|
| ❖ Bachelor of Technology in Electronics and Communication (2018)
Vaagdevi Institute of Technology, Proddatur (Affiliated to JNTUA University) | 66% |
| ❖ Diploma in Electronics and Communication (ECE) (2015)
Government Polytechnic, Kamalapuram | 80.2% |

PERSONAL INFORMATION:

Name:	Arun Kumar G
DOB:	31 July 1996
Marital Status:	Unmarried
Nationality:	Indian
Language Known:	English, Telugu
Address:	Proddatur, Andhra Pradesh.

DECLARATIONS:

I hereby declare that the above written particulars are true to the best of my knowledge and belief.

Place: Bengaluru

Arun Kumar G